Food and Agriculture Organization of the United Nations

ICARDA
Science for resilient livelihoods in dry areas

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Sweden Sverige

Regional Initiative on Water Scarcity

inter-Regional Technical Platform Water Scarcity

WEBINARS SERIES ON RAINWATER HARVESTING

5th October 2022 – 11th October 2022

Module n°7:

Addressing Climate Change Impacts through Rainwater Harvesting

Boost your crop water productivity in selected farming systems: assess and adapt

Session 1: Water buffering for climate change adaptation and biodiversity enhancement

Date: 5th October 2022 **Time**: 13 h 00 – 14 h 30 (GMT+2)

***Register**: Water buffering for climate change adaptation and biodiversity enhancement*

Session 2: Rainwater harvesting for climate change adaptation and ecosystem preservation

Date: 11th October 2022 **Time**: 13 h 00 – 14 h 30 (GMT+2)

***Register**: Rainwater harvesting for climate change adaptation and ecosystem preservation*

BACKGROUND

Water scarcity is a critical challenge in countries and regions with arid and semi-arid conditions and fast-growing water demands. This is mostly the case in all of the countries across the NENA Region, where food demand and food insecurity continue to grow, widening the water supply-demand gap in most countries. Furthermore, the anticipated impact of climate change in the NENA region is indicative towards an increased intensity of rainstorms and flash floods, while the total amount of annual precipitation rates shows declining trends. During the last 20 years, interest has been renewed in rainwater harvesting (RWH) as one of the most effective climate adaptation strategies to cope with water scarcity, thus reducing the pressure on the limited freshwater resources and increasing rainfed crop productivity.

This webinar series on Rainwater Harvesting aims to provide an overview of RWH applications and illustrates, with country examples, the diversity of approaches. It will try to unlock the potential of rainwater harvesting applications in the NENA region by highlighting the role of rainwater management in overcoming the current challenges in achieving water security. It will also shed the light on the need for innovations in economics, social consensus and supportive laws and regulations to ensure a widespread uptake of RWH technologies in the NENA region.

Main Conveners:

FAO/WEPS-NENA project & Water Scarcity Initiative (WSI) with: ICARDA, ICID and FAO inter-Regional Technical Platform on Water Scarcity (*i*RTP-WS)

Contributing Partners:

Arab Centre for the Studies of Arid Zones and Drylands (ACSAD), ESCWA, MetaMeta and University of Natural Resources and Life Sciences

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Languages:

English and French with live translation

Recording:
https://dgroups.org/fao/waterproductivity/rwh

ABOUT THE MODULE 7

There are two webinars available for discussing how rainwater systems can help to address the challenge of climate change. The module will draw on a review of ACSAD experiences in implementing rainwater harvesting projects in the MENA region from a climate change perspective, as well as a review of twenty years of watershed development in semi-arid India. The role of water buffering at catchment level and how water buffering can transform landscapes and livelihoods will be further explored. Discussions will continue with a reflection on the effects of rainwater harvesting on biodiversity enhancement and ecosystem services.

SUGGESTED READINGS

- Batchelor et al. (2003) Watershed development: a solution to water shortages in semi-arid India or part of the problem? https://core.ac.uk/download/pdf/6569947.pdf
- Gaur S. and Milne G. (2015) Improving operational effectiveness and impact of the Integrated Watershed Management Program in India. World Bank Agriculture Global Practice Discussion Paper https://www.profor.info/sites/profor.info/files/Improving%20Operational%20Effectiveness%20and%20Impacts%20of%20the%20Integrated%20Watershed%20Management%20Program%20in%20India_0.pdf
- Haddad, M., Strohmeier, S.M., Nouwakpo, K., Rimawi, O., Weltz, M., Sterk, G. 2022. Rangeland restoration in Jordan: Restoring vegetation cover by water harvesting measures. International Soil and Water Conservation Research. https://doi.org/10.1016/j.iswcr.2022.03.001
- Knoop L., Sambalino F., and F. Van Steenbergen (2012). Securing Water and Land in the Tana Basin: a resource book for water managers and practitioners. Wageningen, The Netherlands: 3R Water Secretariat https://www.hydrology.nl/images/docs/dutch/2012.11_Tana_Basin_Kenya.pdf
- Steenbergen F. van and A. Tuinhof. (2009). Managing the Water Buffer for Development and Climate Change Adaptation. Groundwater recharge, retention, reuse and rainwater storage. Wageningen, The Netherlands: MetaMeta Communications. https://www.bebuffered.com/downloads/3R_managing_the_water_buffer_2010.pdf
- Strohmeier, S., Fukai, S., Haddad, M., Al Nsour, M., Mudabber, M., Akimoto, K., Yamamoto, S., Evett, S., Oweis, T. 2021. Rehabilitation of degraded rangelands in Jordan: The effects of mechanized micro water harvesting on hill-slope scale soil water and vegetation dynamics. Journal of Arid Environments. 185. https://doi.org/10.1016/j.jaridenv.2020.104338
- Tuinhof, A., van Steenbergen, F., Vos, P. and L. Tolk. 2012. Profit from Storage. The costs and benefits of water buffering. Wageningen, The Netherlands: 3R Water Secretariat. https://www.bebuffered.com/downloads/profit-from-storage-reprint-2013_digitalvs.pdf
- The following websites: https://www.bebuffered.com/ ; https://metameta.nl/the-work-we-do/landscape-rehabilitation and https://www.greener.land/

CERTIFICATE OF ACHIEVEMENT

A digital certificate of achievement will be offered by FAO to participants who:

- Attend at least four webinars belonging to Module 1 to Module 4 and five webinars belonging to Module 5 to Module 10;
- Obtain a score of no less than 70 percent in each of two written assessments that include multiple-answer questions and reflect the content of the aforementioned modules.

The first assessment covering the modules 1 to 4 was organized in August 2022. The link to the second assessment, covering the modules 5 to 10, will be sent by email at the end of the year to eligible participants who will be given two weeks to submit their answers. Respondents will have the possibility to make two attempts to answer the assessment questions, and only the highest score will be considered.

Important note: *For participants who wish to obtain a certificate of achievement, to be able to track your attendance, you must register and join the webinars using the* ***same e-mail address.***

Session 1

Speakers and moderator

Frank van Steenbergen

has been involved in international policy initiatives such as the Framework for Action Dialogue on Water Governance Guidelines for Spate Irrigation (FAO) and the 'Water for Development' policies for four governments. He was leading the writing team of the Groundwater Governance Vision and Framework for Action – the latter a joint effort of the FAO, UNESCO, World Bank, GEF and IAH. He recently completed the Water Sector Policy for the Islamic Development Bank and Guidelines on Green Roads for Water for the World Bank. He is working with MetaMeta (www.metameta.nl), a social enterprise dedicated to better water management that won two recent environmental awards.

Charles Batchelor

is an independent consultant based in London. His qualifications include a BSc in physics and a PhD in plant sciences. Charles worked for twenty years at the UK's Centre for Ecology and Hydrology (CEH). Since leaving CEH, Charles has worked for various international agencies, universities and NGOs. His interests include water accounting and auditing, water resources management and Agro-hydrology.

A J (Viju) James

holds a PhD from University College London and is an Environmental Economist with over 35 years of research and work experience in a range of development sectors, including agriculture, irrigation, forestry, watershed, water resource and groundwater management and adaptation to climate change. He has worked in India, Afghanistan, Egypt, Ethiopia, Nepal, Sri Lanka, Tanzania, and Vietnam, for governments and donor-supported development projects. His clients include major funding agencies (e.g., World Bank, UNICEF, UNDP, FAO, ADB), research and development institutions (e.g., IRC, ODI), consulting firms, etc. He has given invited talks and seminars, co-edited or co-authored 4 books, contributed 21 book chapters, published 10 academic journal articles and 12 monographs, and contributed around 20 conference or working papers. He has been an External Examiner to the University of Mauritius (2013–15), and Visiting Professor at the Institute of Development Studies, Jaipur (2013–16).

Somayeh Shadkam

is an irrigation specialist in FAO Regional Office for Asia and the Pacific. Somayeh holds a Ph.D. in Water System and Global Change, a M.Sc. in Irrigation and Drainage Engineering and a Master of Engineering in Civil (Green infrastructure). She has more than 15 years of experience in conducting research and consultancy on global change impacts and adaptation in relation to water, food and natural ecosystems, in different research institutes and international organizations such as IIASA, IWMI and FAO. She has a strong background in hydrological modelling and application of remote sensing data and has extensive field experience. Somayeh is strongly driven by bringing science into practice.

Session 2

Speakers and moderator

Ihab Jnad

is the Director of the Water Resources Department at the Arab Center for the Studies of Arid Zones and Dry Lands (ACSAD) in Damascus, Syria since 2015. He previously served as the head of the Water Resources Development Program at ACSAD from 2002 to 2015. He also currently acts as Associate Professor at the Faculty of Agriculture, Damascus University in Syria since 2001. He specializes in water resources management using mathematical modeling, impact assessments of climate change on water resources, and estimations of crop water requirements from satellite images. His areas of expertise also include rainwater harvesting and irrigation management. He holds a Ph.D in water resources engineering from Texas A&M University, USA.

Stefan Strohmeier

is a Senior Researcher at the University of Natural Resources and Life Sciences (BOKU), Institute for Soil Physics and Rural Water Management (SoPhy) in Vienna, Austria. He holds a PhD in Environmental Engineering and Water Management. Prior to his re-engagement at BOKU University in 2022, he spent the past eight years in the dry regions of North Africa and Near East, East- and Subsaharan Africa and Central Asia. From 2014 to 2022, Stefan held scientific and project managerial positions at ICARDA, Amman office. From 2020 to 2022, he led ICARDA's research sub-team called "Restoration Initiative on Dryland Ecosystems". Stefan's research interests are on sustainable land and water management towards combatting land degradation. Stefan published dozens of SCI publications and book chapters including written publications and presentations provided at international fora.

Tarek Sadek

is a First Economic Affairs Officer (Water Resources and Climate Change) in the Sustainable Development and Productivity Division of ESCWA (Lebanon) since May 2007. He has 25 years of experience in the field of climate change, environmental engineering, water resources management and planning, hydrogeology, water quality and environmental aspects. He has more than 20 official publications in books, international journals and conferences in the field of water resources and environmental management. He holds a Doctorate in Environmental Engineering from the University of Melbourne, Australia, 1998. He completed his Masters in Hydrology from the University College Galway (UCG), Galway, Republic of Ireland, in 1992 and a B.Sc. in Civil Engineering from Cairo University in 1988. From 1999 to 2006, he was the Director of the National Water Resources Plan project – Planning Sector at the Ministry of Water Resources and Irrigation in Egypt.

Module 7: Addressing Climate Change Impacts through RWH

Session 1: Water buffering for climate change adaptation and biodiversity enhancement

Date: Wednesday, 5th October, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	**Welcoming Remarks** Introduction to the session objectives and speakers **By: Somayeh Shadkam,** Irrigation Specialist, FAO
01:05 – 01:30	Presentation on **"Transforming Landscapes to Transform Lives - focus on the 3R Approach"** **By: Frank van Steenbergen,** Director at MetaMeta
01:30 – 01:55	Presentation on **"Watershed development: A solution to water scarcity in semi-arid India or part of the problem?"** **By: Charles Batchelor,** Independent Consultant, and **Viju James**, Environmental Economist
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up

Module 7: Addressing Climate Change Impacts through RWH

Session 2: Water buffering for climate change adaptation and biodiversity enhancement

Date: Tuesday, 11th October, 2022

Time: 01:00 PM- 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	**Welcoming Remarks** Introduction to the session objectives and speakers **By Moderator: Tarek Sadek,** First Economic Affairs Officer (Water Resources and Climate Change), ESCWA
01:05 – 01:30	Presentation on **"ACSAD's Experience in Implementing RWH for Rangeland Rehabilitation and Supplementary Irrigation, in the Arab Region"** **By Ihab Jnad,** Director of the Water Resource Department, ACSAD
01:30 – 01:55	Presentation on **"The effects of RWH on Ecosystem Services"** **By: Stefan Martin Strohmeier,** Senior Researcher at the University of Natural Resources and Life Sciences, Vienna
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up

This activity is implemented within the regional project "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" under the umbrella of the Water Scarcity Initiative. This regional project is implemented by the Food and Agriculture Organization of the United Nations and funded by the Swedish International Development Cooperation Agency (SIDA).

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